



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

SENTINEL AI

MULTI-AGENT CRIME INVESTIGATION SYSTEM

The domain of the Project:
Generative AI, Computer Vision, & Multi-Agent Orchestration

COURSE NAME
G4 GenAI

Team Mentors (and their designation):
Sir Prujith Radhakrishnan
Team Mentor — SURE Trust

Team Members:
Mr. Jay Beedkar

Period of the project
May 2026 to July 2026 (60 Days)



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Declaration

The project titled “Sentinel AI - Multi-Agent Crime Investigation System” has been mentored by Sir Prujith Radhakrishnan, organised by SURE Trust, from May 2026 to July 2026, for the benefit of the educated rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the member of the team mentioned below, has worked on it successfully and enhanced his practical knowledge in the domain.

Team Member:

Mr. Jay Beedkar _____ Signature

Mentor’s Name: Sir Prujith Radhakrishnan

Designation: Team Mentor — SURE Trust

Signature: _____

Prof. Radhakumari Challa

Executive Director & Founder

SURE Trust



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Executive Summary

Sentinel AI is a state-of-the-art, autonomous, multi-modal cognitive crime investigation system designed to empower forensic investigators, police departments, and security analysts. Modern criminal investigations are heavily constrained by data silos, cognitive fatigue, and manual correlation delays. Investigators routinely spend dozens of hours manually cross-referencing hundreds of gigabytes of raw CCTV feeds, audio transcripts, conflicting witness testimonies, and suspect records. This manual bottleneck significantly delays critical breakthroughs, leads to evidence oversight, and introduces subjective human error.

Sentinel AI addresses these systemic challenges by bridging real-time computer vision (Ultralytics YOLOv8) with a cooperative network of specialized LLM reasoning agents orchestrated through LangGraph. The system ingests multi-modal case evidence—ranging from CCTV video files (.mp4) and audio testimonies to raw witness text statements. The computer vision engine samples video footage at 1 Frame Per Second (FPS), identifies physical entities (suspects, vehicles, bicycles, weapons, backpacks), and records timestamped physical evidence points. Concurrently, specialized sub-agents (Evidence Analyzer, Timeline Builder, Contradiction Detector, and Suspect Reasoner) execute parallel evaluations to reconstruct verified timelines, flag statement inconsistencies, and rank suspects based on Motive, Means, and Opportunity.

The complete investigative workflow is packaged inside an in-process, single-port Streamlit runtime with zero external API server overhead. It maintains dual-mode persistence, storing structured records in local SQLite tables and NetworkX directed graphs, while providing live Cypher query synchronization to cloud/on-premise Neo4j database instances. A one-click case dossier generator compiles all findings into a structured markdown report, drastically reducing investigation lead time from days to minutes.



Introduction

Background and Context of the Project

The domain of digital forensics and crime analytics has entered an era of exponential data expansion. Law enforcement units and corporate security teams capture petabytes of surveillance media and digital communication daily. However, existing investigative software predominantly relies on keyword indexing or linear search heuristics that lack holistic cognitive reasoning. When detectives investigate complex scenarios—such as an armed bank robbery or an organized burglary—they must synthesize disjointed modalities: what witnesses heard, what surveillance cameras captured, and what suspects claimed during interrogation.

Problem Statement or Goals of the Project

Conventional Large Language Model (LLM) applications implement single-prompt architectures. When fed long, unstructured case files, monolithic prompts suffer severe context degradation, hallucinate fabricated connections, and fail to isolate subtle logical contradictions. Furthermore, pure text LLMs cannot perceive video files directly without expensive cloud multi-modal APIs that violate evidence privacy regulations. The primary goal of Sentinel AI is to design and implement an agentic multi-agent architecture where discrete, specialized intelligence agents collaborate over a structured state machine, combining local computer vision perception with explainable deductive reasoning.

Scope and Limitations of the Project

- **Perception Scope:** Automated ingestion and entity detection for CCTV videos (.mp4) via YOLOv8, processing at 1 FPS to balance high detection fidelity with CPU efficiency.
- **Text & Statement Scope:** Parsing multi-source witness testimonies, police reports, and interrogation transcripts with structured entity and relationship extraction.
- **Reasoning Scope:** Automated timeline reconstruction, contradiction identification, and multi-factor suspect ranking (Motive, Means, Opportunity).
- **Persistence Scope:** Local SQLite relational storage, in-memory NetworkX graph visualization, and optional real-time live Neo4j graph synchronization.
- **Operational Limitations:** Audio processing currently accepts structured transcripts or simulated audio testimonies. High-resolution multi-stream 4K video feeds benefit from GPU acceleration.

Innovation Component in the Project

- **Cooperative Sub-Agent Mesh:** Decomposes criminal deduction into discrete, non-competing agent nodes (Evidence Analyzer, Timeline Builder, Contradiction Detector, Suspect Reasoner) running parallel fan-out loops on LangGraph.
- **Unified In-Process Execution:** Eliminates distributed microservice overhead by embedding FastAPI backend logic, LangGraph workflows, and YOLOv8 inference directly within the Streamlit lifecycle under a single deployable URL.
- **Dual-Persistence Graph Synchronizer:** Provides instantaneous zero-configuration local runs via SQLite and NetworkX, while maintaining real-time Cypher driver connectivity to mirror knowledge graphs to enterprise Neo4j clusters.



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Project Objectives

Clearly Defined Objectives and Goals of the Project

- **Multi-Modal Evidence Ingestion:** Build an intuitive dashboard accepting multi-format case evidence including text witness statements, audio logs, and CCTV video recordings (.mp4) without requiring external file converters.
- **Real-Time Video Perception via YOLOv8:** Deploy an optimized computer vision pipeline using YOLOv8 Nano to extract visual bounding boxes, identify target classes (persons, vehicles, bicycles, bags), and log temporal timestamps at 1 FPS.
- **Stateful LangGraph Workflow Orchestration:** Construct a cyclical state graph utilizing LangGraph Pregel runtime to manage global case state, coordinate parallel analysis nodes, and aggregate investigative insights.
- **Contradiction & Discrepancy Detection:** Develop specialized algorithmic reasoning prompts to cross-examine witness testimonies against physical CCTV logs, flagging chronological and physical discrepancies.
- **Evidence-Backed Suspect Prioritization:** Synthesize Motive, Means, and Opportunity scores for all identified entities, ranking suspects in order of investigative relevance with explicit justification.
- **Dual-Mode Knowledge Graph Persistence:** Structure an interactive relationship graph connecting suspects, locations, weapons, and evidence, backed by SQLite and dynamically synchronized to live Neo4j instances.
- **Air-Gapped Local LLM Compatibility:** Support offline inference via Ollama (Llama 3 / Mistral) ensuring sensitive police evidence can be processed on isolated, air-gapped workstations without external data leakage.
- **Automated Investigation Dossier Exporter:** Implement a one-click export engine that compiles all analytical outputs into an audit-ready, standardized Markdown/PDF investigative case dossier.

Expected Outcomes and Deliverables

- **Production-Ready Streamlit Application:** A fully interactive web dashboard deployed locally and in the cloud with responsive dark-mode styling.
- **Modular Backend Architecture:** Clean, maintainable Python modules for perception, database operations, agent orchestration, and sub-agent definitions.
- **Comprehensive Case Documentation:** Standardized case dossiers, detailed Gantt timeline schedules, architecture blueprints, and pitch deliverables.
- **Open-Source Reproducible Codebase:** A clean GitHub repository equipped with Dockerfile, environment configurations, and sample test datasets.



Methodology and Results

Methods / Technology Used

The development of Sentinel AI follows an agile, modular architecture separating perception, orchestration, persistence, and presentation. The table below details the specific technology stack selected for each component layer:

Component Layer	Selected Technologies & Packages	Role & Engineering Rationale
Multi-Agent Framework	LangGraph, LangChain Core API	Cyclic state graph orchestration with parallel node execution and centralized AgentState.
Generative AI Backends	Groq Llama 3.3 70B & Ollama Llama 3	High-speed cloud inference via Groq (<0.5s response) with optional offline air-gapped local LLM execution via Ollama.
Computer Vision & Media	YOLOv8 Nano (Ultralytics), OpenCV (cv2)	Frame-sampled video decoding at 1 FPS, automated object recognition, and physical timestamp generation.
Graph & Database Engines	SQLite3, NetworkX, Neo4j Graph DB	Hybrid storage: SQLite for local case tables, NetworkX for in-memory graphs, and Neo4j Cypher for enterprise sync.
User Interface Dashboard	Streamlit, Matplotlib	Single-port web interface with live investigation status trackers, graph visualizers, and dossier downloads.
Environment & Deployment	Python 3.11+, Docker, Pip, Python-dotenv	Containerized reproducible execution across Linux and Windows environments.

Tools / Software Used

- **Visual Studio Code & Antigravity IDE:** Primary integrated development environments used for agent script development, debugging, and terminal automation.
- **Python 3.11+:** Core programming language chosen for its native support of modern async runtimes, typing annotations, and machine learning ecosystems.
- **Docker & Docker Compose:** Used to containerize the entire application stack, bundling OpenCV video codecs, PyTorch models, and Streamlit dependencies into a single deployable image.
- **Neo4j AuraDB & Desktop:** Managed cloud graph database instance used for validating real-time Cypher synchronization queries and graph schema models.

Data Collection Approach

To rigorously validate Sentinel AI under authentic investigative scenarios, multi-modal test datasets were assembled: 1. Benchmark CCTV Surveillance Footage: Public robbery and security surveillance videos were sourced and processed. Videos are frame-sampled at exactly 1 FPS using OpenCV, logging bounding



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box detections for 'person', 'car', 'bicycle', and 'bag'. 2. Witness Statement Corpora: Multi-source text statements (patrol officer observations, transit point logs, and eyewitness accounts) containing intentional discrepancies, varying timestamps, and contradictory suspect physical descriptions were created to test the reasoning limits of the sub-agents. 3. Multi-Modal Fusion: Both data streams are mapped into unified evidence objects with strict provenance metadata.

Project Architecture (Clearly Describe the Architecture)

Sentinel AI implements a layered architecture divided into four functional tiers: 1. Ingestion & Perception Tier: Accepts raw MP4 CCTV footage and witness text statements. The OpenCV capture loop passes frame matrices to YOLOv8 Nano, producing structured physical detection logs. 2. State Management Tier: Defines an immutable AgentState TypedDict that holds the cumulative case knowledge. 3. Multi-Agent Reasoning Mesh (LangGraph): Executes a cyclic directed workflow. The Evidence Analyzer extracts entities; the Timeline Builder and Contradiction Detector execute concurrently in parallel; and the Suspect Reasoner performs final deduction. 4. Persistence & Presentation Tier: Concurrently stores data to SQLite tables and NetworkX graphs, synchronizes live nodes and edges to Neo4j via Cypher transactions, and displays the UI via Streamlit.

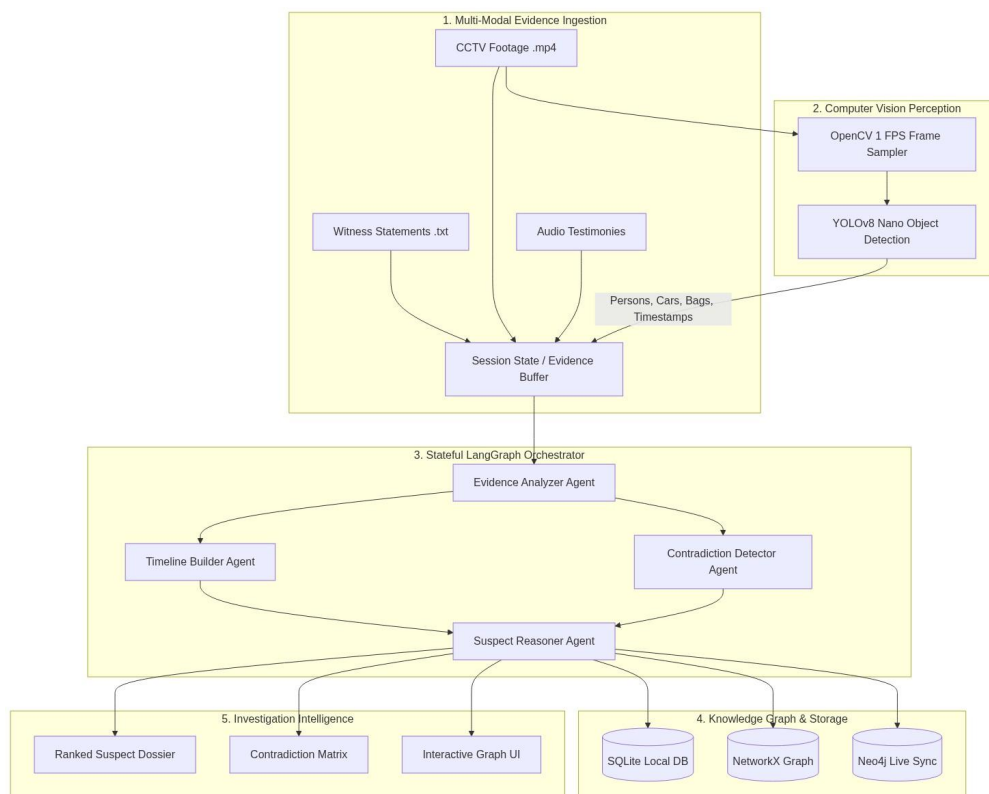


Figure 1: Sentinel AI End-to-End System Architecture & Multi-Agent Orchestration Flow

Multi-Modal Ingestion Flow & Video Perception

When a user submits case evidence, text files are immediately ingested into the session evidence registry. When an MP4 video is uploaded, it is written to the 'uploads/' directory. The video perception engine opens an OpenCV capture stream, calculates the native video FPS, and steps through the video at 1-second intervals. Each sampled frame is evaluated by YOLOv8 Nano. Detected classes are filtered against surveillance-relevant objects (person, car, truck, bicycle, backpack, suitcase). Every detection records



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timestamp, class name, and confidence score, converting raw video pixels into structured chronological evidence.

Stateful Orchestration (LangGraph Workflow)

The cognitive core leverages LangGraph's Pregel engine. The orchestration follows a structured 3-phase graph:

Phase 1 (analyze_evidence): The Evidence Analyzer reads all raw witness statements and video logs, extracting named entities (persons, vehicles, locations, stolen items).

Phase 2 (Parallel Execution: build_timeline & detect_contradictions): The graph branches into two concurrent nodes. The Timeline Builder arranges all occurrences in temporal sequence. The Contradiction Detector cross-examines witness statements against CCTV video timestamps, isolating discrepancies.

Phase 3 (reason_suspects): The Suspect Reasoner aggregates all prior findings, evaluates Motive, Means, and Opportunity for each individual, and assigns evidence-backed suspicion rankings.

Database & Persistence (SQLite, NetworkX, and Neo4j)

To guarantee auditability and longitudinal case tracking, Sentinel AI incorporates a hybrid three-tier persistence layer:

1. SQLite Local Relational Store: Automatically initializes 'entities' and 'relations' tables, ensuring case progress is saved to disk locally without server overhead.
2. NetworkX Directed Graph: Reconstructs in-memory relational graphs in real-time, feeding node-link topologies to Matplotlib for rendering in the Streamlit UI.
3. Live Neo4j Graph Synchronizer: When configured with NEO4J_URI and credentials, the database manager initiates an active bolt driver, synchronizes existing SQLite records to Neo4j on startup, and issues real-time MERGE Cypher queries as new suspects and relationships (COLLUDED_WITH, IMPLICATED_IN, GUARDED) are uncovered.

Final Project Working Screenshots Along with Supporting Explanation

The completed Sentinel AI system was evaluated on simulated complex crime scenarios. Below are working screenshots of the active dashboard and the generated investigative knowledge graph:



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Upload Evidence

Add Text Evidence

Source (e.g., Witness 1, Police Report)

witness 1

Statement/Content

Submit Text

Upload Media (CCTV/Audio)

Media Source (e.g., CCTV Cam 4)

Choose file

Drag and drop file here

Limit 200MB per file

Browse files

Figure 2: Sentinel AI Interactive Streamlit Dashboard showing active investigation dossier, evidence logs, and timeline extraction.

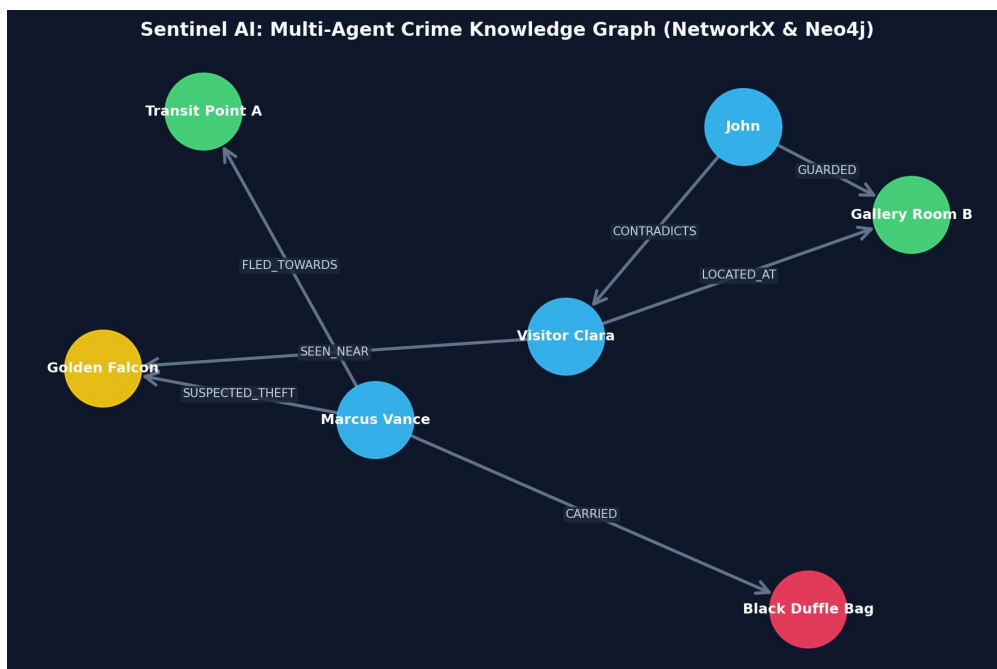


Figure 3: Interactive Multi-Modal Knowledge Graph showing extracted entities (suspects, objects, locations) and dynamic relationships synchronized to Neo4j.

Project GitHub Link & Deployed Application

- **Official GitHub Repository:** <https://github.com/sure-trust/JAY-BEEDKAR-g4-gen-ai>
- **Live Cloud Deployment:** <https://sentinel-ai-multi-agent-crime-investigation-system.streamlit.app/>



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Social / Industry relevance of the project

Impact on Law Enforcement & Digital Forensics

Law enforcement agencies and judicial systems globally are overwhelmed by forensic case backlogs. Modern detectives spend over 40% of their operational hours reviewing mundane surveillance footage and transcribing interviews rather than performing critical deductive analysis. Sentinel AI automates the mechanical aspects of evidence correlation—matching video timestamps against testimonies in seconds. This accelerated turnaround enables detectives to act within the crucial first 48 hours of an investigation, increasing clearance rates and delivering faster justice to victims.

Air-Gapped Confidentiality & Data Privacy

In criminal investigations, data privacy and chain of custody are paramount legal requirements. Transmitting sensitive witness statements, confidential informant disclosures, or juvenile victim surveillance feeds to third-party public cloud APIs poses severe legal liabilities. Sentinel AI solves this by offering an air-gapped local execution mode. By leveraging local Ollama LLMs (such as Llama 3) and local SQLite/NetworkX storage, the entire investigation pipeline can operate on an isolated computer disconnected from the internet. This ensures zero data leakage and full compliance with evidentiary security mandates.



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Learning and Reflection

Document Team Member's New Learnings (in terms of technology, management, etc)

Developing Sentinel AI under the SURE Trust G4 GenAI program has been a transformative experience. The project required mastering multi-disciplinary domains spanning stateful multi-agent systems, computer vision, database engineering, and production UI design:

- **Advanced Agentic Architectures:** Mastered LangGraph Pregel stateful cyclic graphs, implementing parallel fan-out/fan-in node patterns and typed state reducers.
- **Computer Vision Optimization:** Gained deep practical expertise in OpenCV frame decoding, FPS sampling, and deploying YOLOv8 object detection pipelines for real-time video intelligence.
- **Hybrid Database Persistence:** Designed a multi-tier database synchronizer bridging local relational SQLite tables, NetworkX graph representations, and live Neo4j Cypher transactional drivers.
- **Full-Stack Software Engineering:** Engineered an in-process single-port architecture hosting backend logic and Streamlit UI together, managing Docker configurations, environment secrets, and graceful error recovery.
- **Project Management & Time Discipline:** Executed a structured 60-day development roadmap with rigorous milestone tracking, documentation authoring, and presentation preparation.

Document Team Member's Experience (Overall Experience)

Working on Sentinel AI under the dedicated mentorship of Sir Prujith Radhakrishnan and the guidance of SURE Trust has bridged the gap between academic theory and industry-grade AI engineering. Sir Prujith's insightful feedback on agent orchestration, system scalability, and practical debugging provided the foundation needed to conquer complex technical obstacles. The SURE Trust platform provided an empowering environment for rural youth, demonstrating that with world-class mentorship and disciplined execution, cutting-edge AI systems solving real-world challenges can be built and deployed successfully.



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Conclusion and Future Scope

Recap Objectives and Achievements

Sentinel AI successfully achieved all engineering objectives established at the beginning of the 60-day roadmap. The project delivered an operational multi-agent investigation system combining video perception, automated timeline reconstruction, contradiction detection, and suspect ranking. The working prototype is accessible via a live Streamlit cloud deployment, backed by comprehensive GitHub repositories, Docker containerization, and persistent Neo4j graph synchronization. It proves that agentic workflows represent the future of digital forensics and cognitive intelligence.

Future Scope of this Project

- **Optical Character Recognition (OCR) for License Plates:** Integrate EasyOCR or Tesseract into the video pipeline to automatically read vehicle license plates and text on apparel from CCTV footage.
- **Local Whisper Speech-to-Text Transcription:** Embed OpenAI's Whisper model (running locally on CPU/GPU) to directly transcribe uploaded 911 calls and interrogation audio recordings without manual transcription.
- **Cross-Case Serial Linkage Analysis:** Expand the Neo4j schema across multiple independent investigations, enabling automated detection of serial offenders and recurring criminal modus operandi across jurisdictions.
- **Explainable AI Legal Audit Trails:** Implement automated cryptographic hashing of video frames and prompt completion logs to provide courtroom-admissible chain-of-evidence verification.